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Origin of hematite carrying a Late Paleozoic remagnetization in a quartz sandstone bed from the Silurian Rose Hill Formation, Virginia, USA

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Abstract

Paleomagnetic and petrographic studies of a quartz sandstone bed from the Silurian Rose Hill Formation of the Central Appalachians show that hematite carrying a Late Paleozoic remagnetization is authigenic. Comparison of paleopole positions from the Rose Hill Formation with Late Paleozoic poles for North America suggests a synfolding remagnetization. The sandstone cement is characterized by pore-lining hematite, pore-filling berthierine and intergrowths of hematite, berthierine and quartz. These cements postdate pressure solution of detrital quartz grains. Late Paleozoic cementation is a consequence of diagenesis associated with Alleghanian deformation. Detrital Fe-Mg silicate grains are believed to have been a local source of Fe for some authigenic hematite and berthierine, although mass balance calculations suggest that an external source is also required. Some berthierine aggregates are pseudomorphic after detrital Fe-bearing silicate grains. Trace amounts of detrital tourmaline are the only remnants of the detrital Fe-Mg silicates.

1. Introduction

Widespread Late Paleozoic remagnetization of sedimentary rocks in eastern North America is well established [1,2]. The remagnetization is characterized by SSE declinations and shallow inclinations. Chemically active orogenic fluids are believed to have played a role in the authigenesis of the magnetic minerals and remagnetization [e.g., 3,4]. Hematite authigenesis is considered to be responsible for the Late Paleozoic remagnetization of many Appalachian red beds, including the Upper Ordovician Juniata Formation [5,6], the Middle Silurian Rose Hill Formation [7,8], the Clinton ironstones [9], the Devonian Catskill Formation [10] and the Mississippian Mauch Chunk Formation [11]. Both Alleghanian and pre-Alleghanian magnetization components are present in some Appalachian red beds [e.g., 5,6,11] whereas only Late Paleozoic remagnetization is preserved in others [e.g., 10]. Synfolding remagnetization of these Appalachian red beds suggests a relationship between remagnetization and deformation. The ages of remagnetization in the Appalachians are older in the south than in the north, which is consistent with the south to north progression of Alleghanian deformation [12]. From blocking temperature–relaxation time relations for hematite [13] and temperatures inferred from thermal maturity indicators, remag-

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netization of Appalachian red beds was not thermally activated [14]. This result supports a chemical remanent magnetization (CRM) mechanism. The age of remagnetization of the Rose Hill Formation is Late Carboniferous to Early Permian, because the paleopole of the Rose Hill Formation lies on this segment of the North American apparent polar wander path [7,8,15]. Comparison of the remagnetized hematite-bearing red sandstones with unremagnetized gray sandstones suggests that a CRM associated with hematite authigenesis is present in the remagnetized rocks of the Rose Hill Formation [8].

The origin of hematite in red beds has implications for the conditions of deposition and diagenesis and the nature of the remagnetization process. The genesis of hematite pigment in red beds has long been debated. Krynine [16] thought that the pigment was detrital, directly eroded from red tropical soils (latosoils). Van Houten [17,18,19] believed that the pigment formed diagenetically by 'aging' of detrital oxides. Walker et al. [20,21,22] proposed that pigmentary hematite is the diagenetic product of long-term intrastratal alteration of detrital ferruginous grains, such as Fe-bearing silicates [e.g., 21,23]. Creer [24] was the first to propose a mechanism for Late Paleozoic remagnetization of North American red beds. He believed that the low latitudes of Laurasia in the Late Paleozoic favored oxidation of Fe-bearing minerals in many Lower/Middle Paleozoic rocks, and that the formation of authigenic hematite was responsible for remagnetization.

Many workers have studied depositional environments of Silurian marine red beds in the Appalachians. Ziegler and McKerrow [25] believed that the Clinton Group represents an eastward transgression during the Middle Silurian over the uplifted landmass formed by the Taconic orogeny. Hunter [26] and Meyer et al. [27] attributed the high Fe content of the Clinton Group to a favorable marine depositional environment. There are different interpretations for the genesis of the hematite in the red beds of the Clinton Group. Cotter and Link [28,29] showed that hematite in the oolitic ironstones of the Clinton Group was a product of diagenetic alteration from chamositic clay mineral precursors. Folk [30] interpreted the authigenic hematite in the hematitic sandstone member of the Rose Hill Formation to have

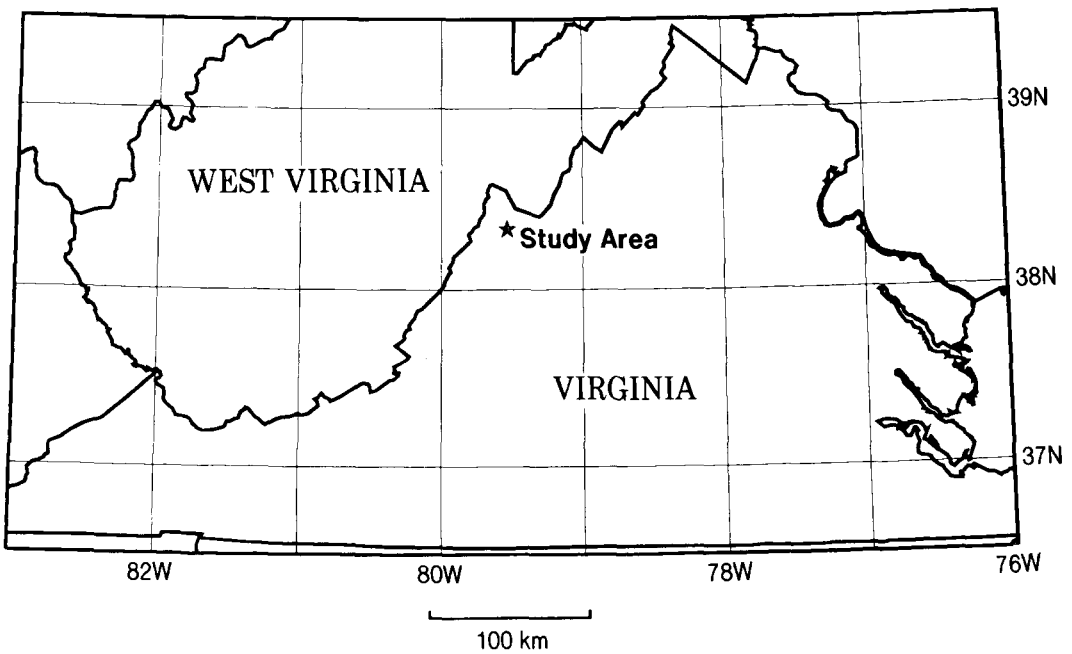


Fig. 1. Map of Virginia and adjacent states showing the sampling locality.

formed by diagenetic alteration of detrital Fe-bearing silicates and placer heavy minerals.

Although the timing of remagnetization in the Rose Hill Formation has been determined to be Alleghanian [7,8,15,9], the diagenetic events associated with hematite authigenesis have not been petrographically linked to remagnetization in these rocks. We present the results of paleomagnetic, petrographic and mineral chemical analyses for a sandstone bed in the Rose Hill Formation to characterize the origin of the hematite responsible for the Late Paleozoic remagnetization, and to determine the diagenetic relationship between the hematite and other minerals.

2. Methods

Oriented hand samples of the Rose Hill Formation of the Silurian Clinton Group were collected from the southeastern limb of the Wills Mountain Anticline in the Appalachian Valley and Ridge Province, Virginia (sample location 38°25′N, 79°35′W, on US Highway 250) (Fig. 1). Samples of dark red sandstone were taken from two beds 3 m apart. No graded bedding, marine fossils or shale interbeds were observed in the samples or in outcrop. The hand samples were then drilled and cut into 23 standard 2.2 cm long core specimens. Stepwise thermal demagnetization was performed using a three-zone furnace constructed with Lindberg components and contained in three nested mu-metal shields. Remanent magnetization was measured with a three-axis superconducting magnetometer manufactured by CTF Systems. The demagnetization data were plotted on orthogonal diagrams [31]. The directions of component magnetizations were determined with principal component analysis [32]. Isothermal remanent magnetization (IRM) acqui-

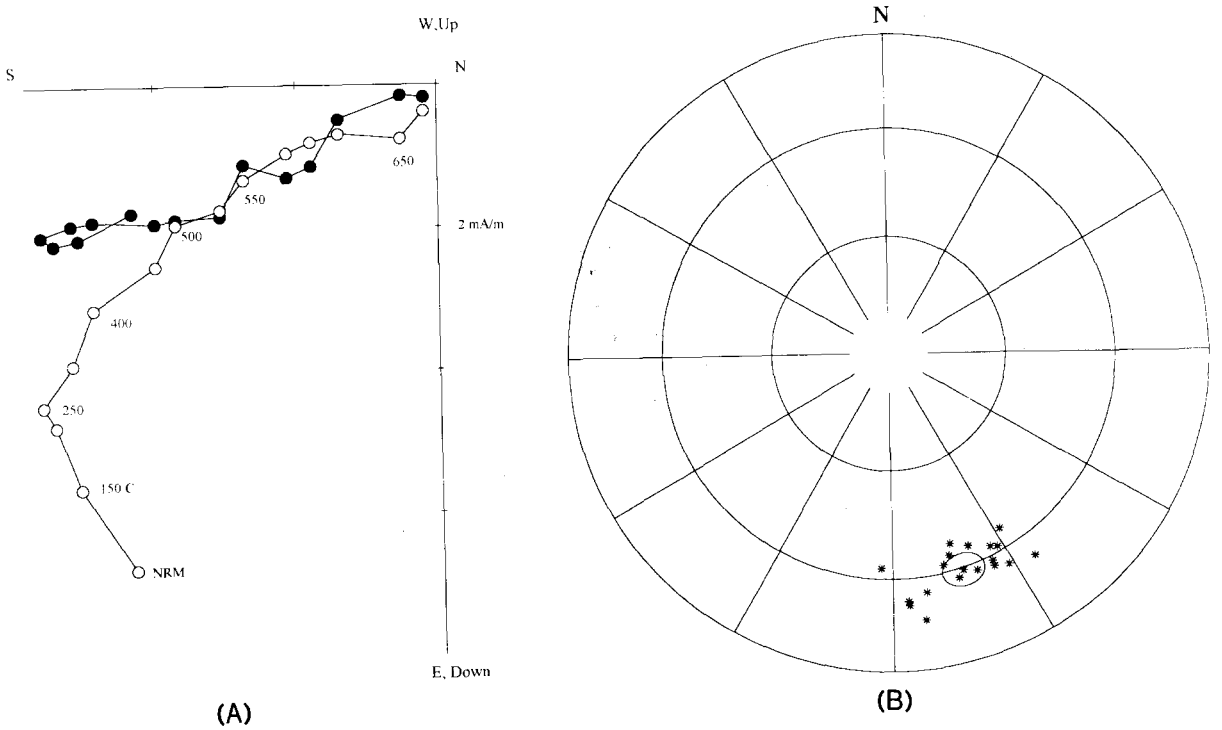


Fig. 2. Paleomagnetic data. (A) Representative orthogonal demagnetization diagram in field coordinates. ○ = vertical projection; ● = horizontal projection. Numbers in the diagram are temperatures (°C) during stepwise thermal demagnetization. (B) Specimen characteristic magnetization directions before structural correction with the cone of 95% confidence (α_{95}) about the mean direction. Lower hemisphere equal-area projection.

Table 1
Summary paleomagnetic and rock magnetic data

Mean Directions (D/I) (field coord.)	(bedding coord.)	N/N_O	k	α_{95}	NRM (mA/m)	MS(10^{-6}) (emu/Oe*cc)	IRM (mA/m)	SKC (mA/m)
162.0/29.2	158.2/-9.2	18/23	53.5	4.8	8.968	30.587	18664.0	5.45

N/N_o = number of specimen directions used compared with the number of specimens demagnetized; MS = magnetic susceptibility; IRM = remanent magnetization after IRM treatment at 300 mT; SKC = size of characteristic component. All data are mean values.

sition was performed to aid in magnetic mineral identification.

To characterize hematite and associated diagenetic phases, we examined thin sections using a petrographic microscope in transmitted and reflected light, and using a scanning electron microscope (SEM) in both secondary electron and backscatter modes. Quantitative analysis of cement and mineral aggregates was performed with wavelength-dispersive spectrometry (WDS) using

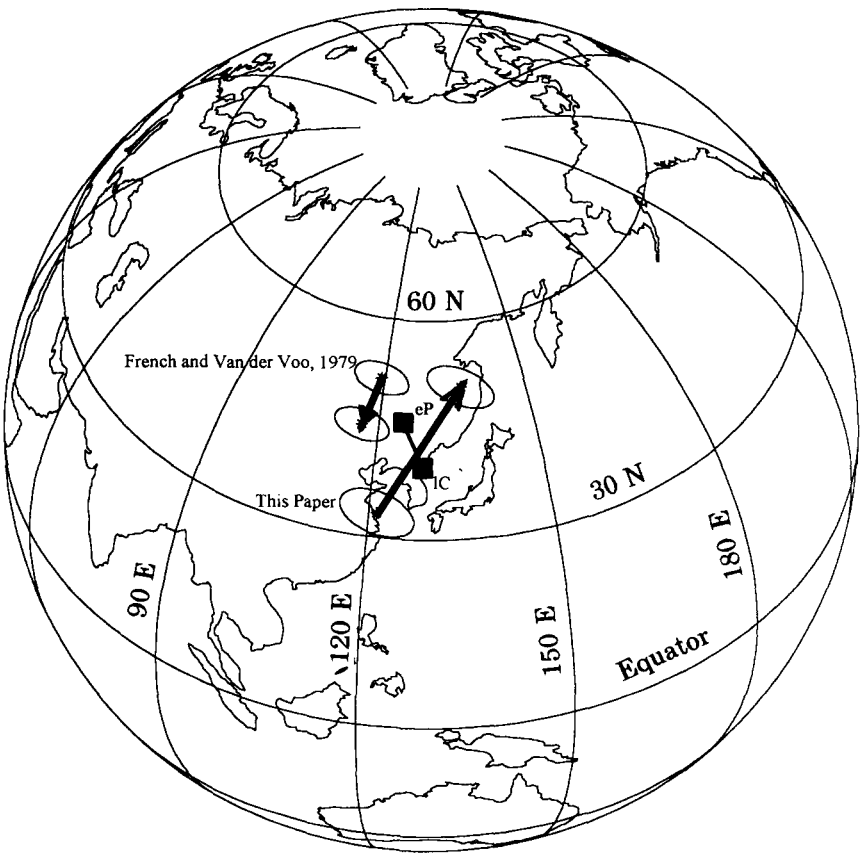


Fig. 3. Comparison of paleopole positions. Arrows connecting pole pairs are tilt correction paths from uncorrected to fully tilt corrected coordinates. Asterisk = pole positions, surrounded by ellipses of 95% confidence (d_P,d_M). Tilt correction paths are shown for the results of this study, and for red bed samples from an earlier paper (remagnetized red bed samples only, from table 1 in [8]). *lC* and *eP* (■) = reference poles for North America for Late Carboniferous and Early Permian respectively [33].

an automated JEOL 733 electron microprobe. WDS analyses were done at a 15 kV accelerating potential and with a 5 nA sample current. The electron beam was focused to a 10 μm spot. Atomic proportions of the structural formulae of berthierine were calculated on the basis of 14 oxygens. Mineral concentrations were estimated from image analysis based on gray levels of backscattered images. Samples of the < 2 μm size fraction were separated by the settling velocity method and analyzed with an automated Phillips X-ray diffractometer (XRD) using Cu Kα radiation.

3. Results

Stepwise thermal demagnetization yields two components of magnetization on orthogonal diagrams (Fig. 2A). The maximum unblocking tem-

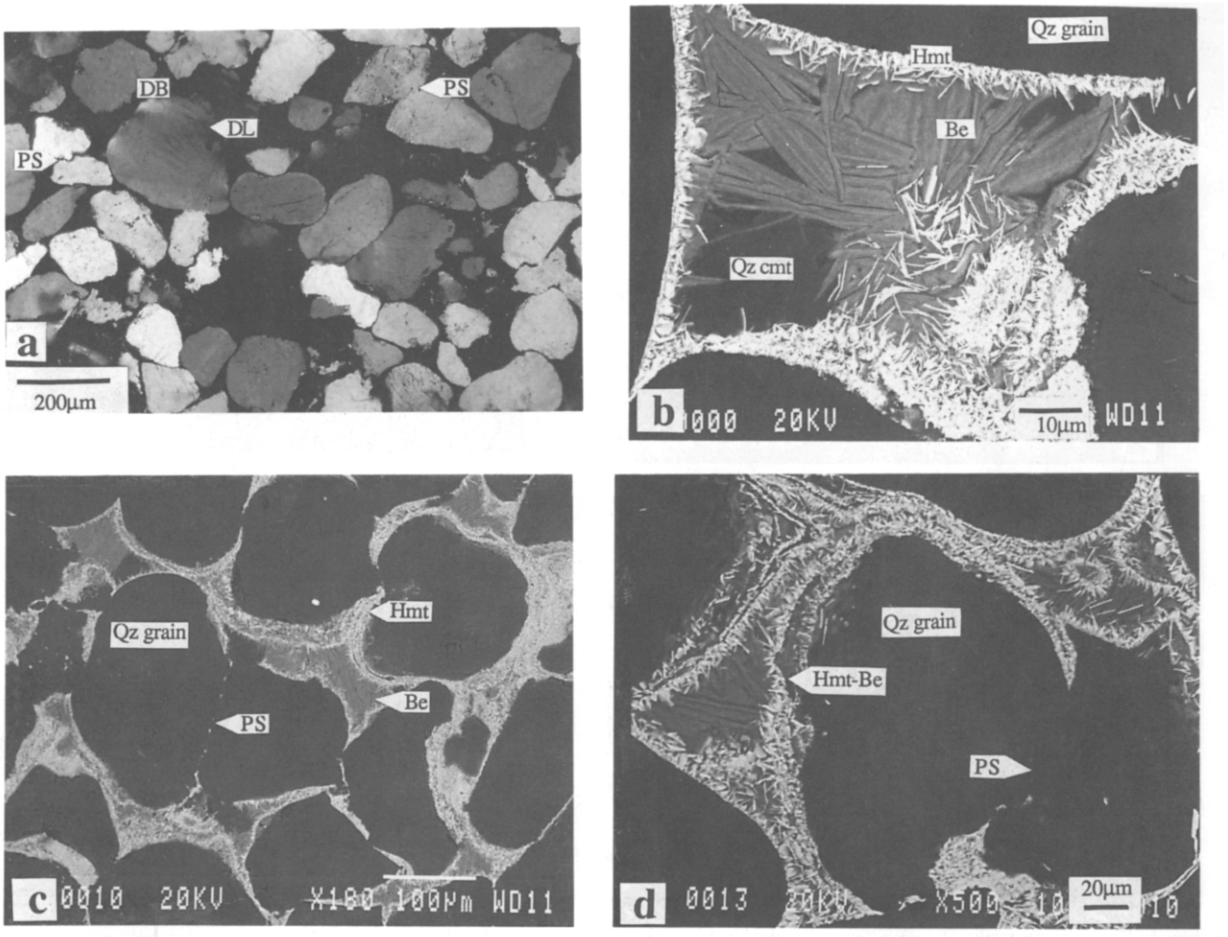


Fig. 4. Photomicrographs showing characteristics of quartz grains and cements. (a) Hematite–berthierine-cemented sandstone with subrounded quartz grains. Quartz grains exhibit pressure solution (*PS*), undulatory extinction, patchy extinction, deformation lamellae (*DL*) and deformation bands (*DB*). Hematite and berthierine cements occur between the quartz grains. (B) Pore spaces are occupied by pore-lining euhedral flaky hematite (*Hmt*), undeformed and randomly oriented laths of pore-filling berthierine (*Be*) and authigenic quartz (*Qt cmt*) in the interior of the pore space. (c) and (d) Pore-lining hematite (*Hmt*) crystallites are oriented mostly perpendicular to the quartz grain boundaries. Multiple intergrowths of hematite–berthierine cements (*Hmt–Be*) occur in pore spaces. Also note quartz grains with sutured grain boundaries (*PS*). (a): Transmitted light with crossed polars. (b)–(d): Backscattered electron images.

perature for the characteristic component is 680°C, indicating that the magnetization resides in hematite. A northerly and down low-unblocking temperature component is interpreted to be a recently acquired magnetization. All the samples have a stable and well-defined characteristic magnetization with SSE declinations and shallow inclinations (Figs. 2A and B, Table 1), which is in agreement with a Late Paleozoic direction rather than a Silurian direction. Because there is little difference in the bedding attitudes of our samples, an incremental fold test cannot be performed. However, comparison between Late Paleozoic poles for North America [33] and the poles obtained from this study provides a constraint on the timing of the magnetization. The area between the tilt-corrected and uncorrected mean poles, representing the pole after partial unfolding, coincides with the expected Late Paleozoic pole position of North America (Fig. 3). These observations are most consistent with a synfolding remagnetization.

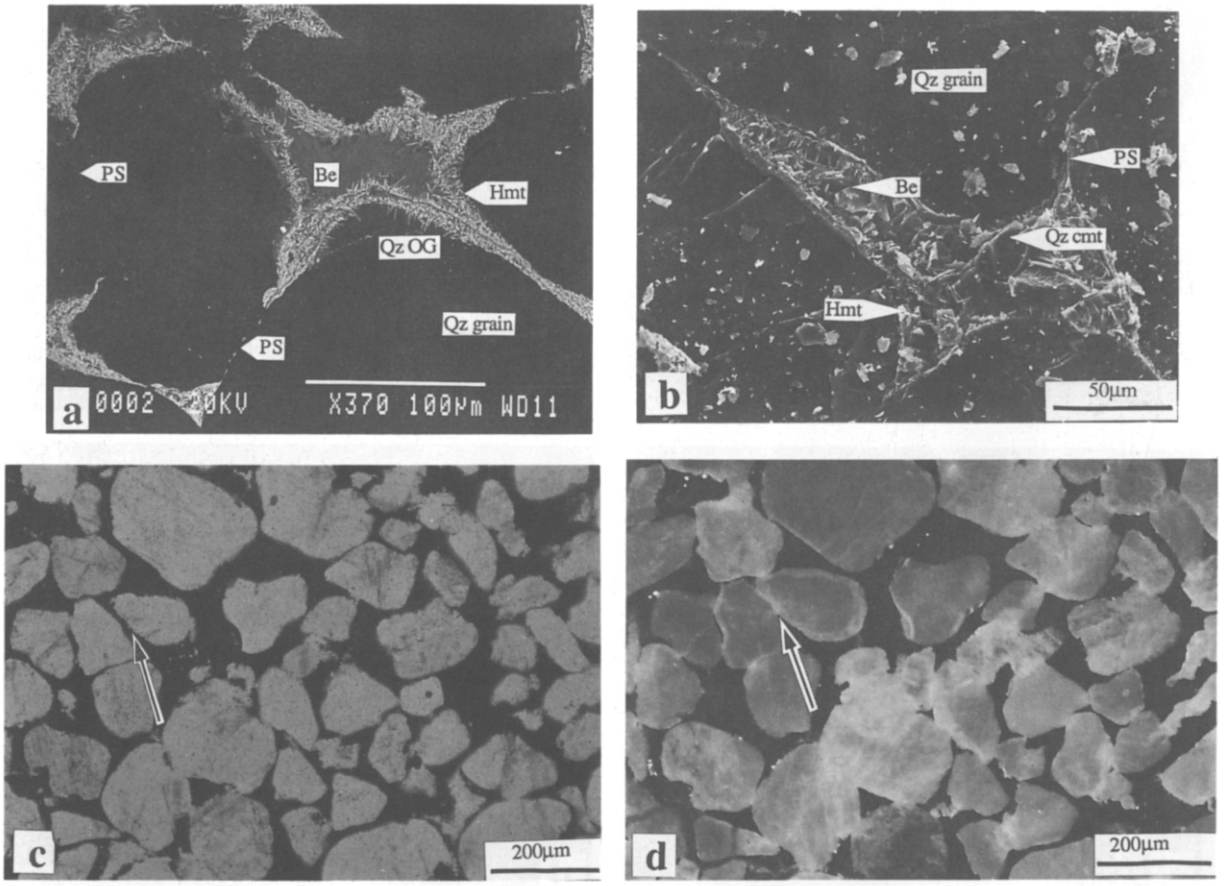


Fig. 5. Photomicrographs showing textural relationships between hematite–berthierine–quartz cements and quartz grain boundaries with pressure solution features. (a) Quartz grains exhibit pressure solution (*PS*) and overgrowths (*Qz OG*). Note that pressure solution seams are terminated adjacent to sutured quartz grain boundaries. (b) Absence of deformation in hematite–berthierine–quartz cements in a pore space between compacted quartz grains with pressure solution (*PS*) at the grain boundaries. (c) and (d) Comparison of transmitted plane polarized light (c) and cathodoluminescence image (d) showing grain boundaries and hematite cement. Note that (d) reveals the true grain contacts with sutured boundaries, which are obscured by overlapping hematite in (c). Arrows indicate that two grains in contact with minor pressure solution in (d) appear to lack contact in (c). The lighter rim of the quartz grain in (d) is the part covered by surficial hematite. (a): Backscattered electron image. (b): Secondary electron image. (c): Transmitted plane polarized light. (d): Cathodoluminescence image.

The sandstone samples of the Rose Hill Formation are dominated by rounded to subrounded detrital quartz grains. Pressure solution is indicated by sutured and interpenetrated grain boundaries. Because sutured grain boundaries were observed in thin sections which were cut parallel to the bedding, this pressure solution feature is possibly due to layer parallel shortening. In addition, quartz grains show undulatory extinction, patchy extinction, deformation lamellae and deformation bands (Fig. 4a). These microstructures are similar to those in the underlying Tuscarora Sandstone [34], although the preferred orientation of the deformation features is not as pronounced. Following Onasch and Dunne [34] and the studies they cited, we ascribe deformation lamellae and deformation bands to layer-parallel shortening prior to Alleghanian folding.

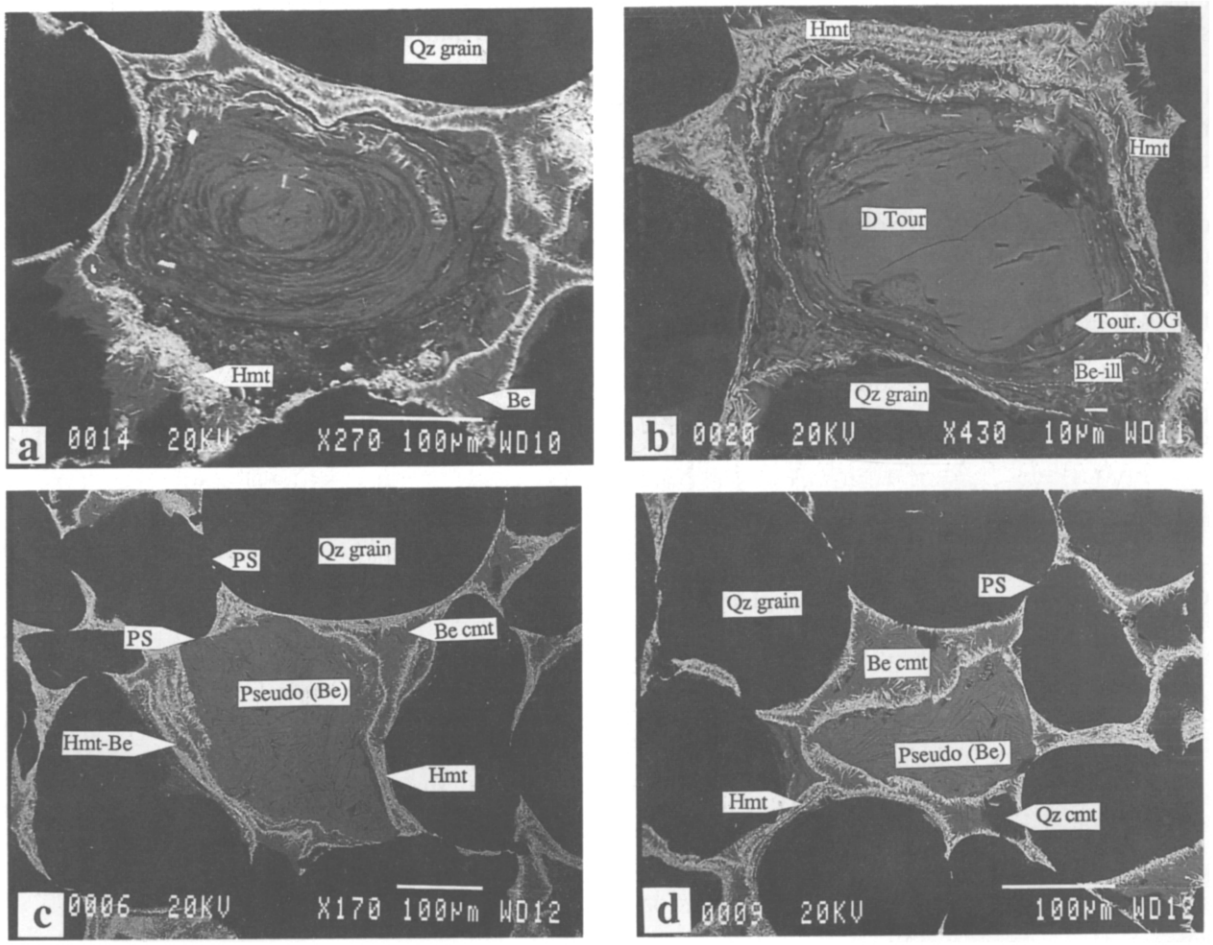


Fig. 6. Backscattered electron images of other petrographic features. (a) An oolitic grain with concentric layering. Oolitic grains are extremely rare (< 0.2% based on point counting) in the samples. The pore spaces between ooids and quartz grains are occupied by hematite–berthierine–quartz cements. (b) A detrital tourmaline (*D Tour*) grain rimmed by concentric intergrowths of hematite (*Hmt*) and berthierine–illite (*Be–ill*). Diagenetic overgrowths (*Tour. OG*) of fine acicular crystals of tourmaline hemimorphically nucleate at positive *c*-crystallographic axis of the detrital tourmaline grain. (c) and (d) Pseudomorphs of berthierine aggregates (*Pseudo (Be)*) with sizes similar to the adjacent compacted quartz grains. Refer to Table 2 for berthierine composition and structural formula of the pseudomorphs. Minor pressure solution (*PS*) occurs at the contact between the berthierine aggregates and quartz grains. Note the pore-lining distribution of hematite (*Hmt*) around the pseudomorphs and the hematite–berthierine intergrowths (*Hmt–Be*) in multiple generations in the adjacent cement.

Detrital framework quartz grains are cemented by pore-lining flaky hematite (5–30 μm), randomly oriented laths of pore-filling berthierine (10–40 μm in the long dimension) and authigenic quartz (20–70 μm) (Fig. 4b). Berthierine cement was identified from the diagnostic 7 Å peak on X-ray diffraction patterns. A very small 14 Å peak relative to the 7 Å peak shows that only a small amount of Fe-rich chlorite is present. Our previous study indicated the presence of minor amounts of Fe-rich illite [35]. Image analysis of representative samples indicates that there is 69% quartz grains, 10% hematite, 11% berthierine and Fe-rich chlorite and 10% other minerals (e.g., tourmaline, illite). Most of the hematite and berthierine have subhedral to euhedral habits. The pore-lining hematite is generally oriented such that its long dimension is perpendicular to the quartz grain boundaries (Figs. 4b, c and d). In addition, multiple generations of hematite–berthierine intergrowths in the cements (Figs. 4c and d) were observed. Deformation features such as pressure solution, deformation lamellae, undulatory extinction and deformation bands are present in the framework quartz grains (Figs. 4a, c and d) and not in the hematite, berthierine and quartz cements (Fig. 4b). Image analysis shows that the present porosity is $< 1\%$, indicating that most of the former pore space is occupied by hematite, berthierine and quartz cements.

Pressure solution features do not extend from the sutured or interpenetrated boundaries between detrital quartz grains into the hematite–berthierine cements (Figs. 5a and b). Deformation features such as grain extrusion, deformation lamellae and kinking are not present in cements adjacent to pressure solution boundaries (Figs. 4a, 5a and 5b). All these observations suggest that hematite–berthierine cements postdate pressure solution of the quartz grains. Less commonly, minor amounts of hematite occur along sutured quartz grain boundaries (Figs. 4c and 5a). Cathodoluminescence images (compare Figs. 5c and d) are much better than transmitted light and SEM images in defining the true grain boundaries. Cathodoluminescence images reveal the true grain contact with the sutured boundaries (Fig. 5d), which are obscured on the transmitted light photomicrograph by hematite cement (Fig.

Table 2
Chemical composition and structural formulae of berthierine

Oxides	Pseudo-1 (n=3)	Cement-1 (n=6)	Pseudo-2 (n=5)	Cement-2 (n=5)
SiO_2	24.55 (±1.19)	25.96 (±0.56)	24.83 (±0.77)	26.45 (±0.09)
Al_2O_3	20.85 (±0.38)	22.06 (±0.29)	23.07 (±0.41)	22.65 (±0.57)
TiO_2	0.01 (±0.01)	0.01 (±0.01)	0.01 (±0.01)	0.01 (±0.01)
FeO	36.14 (±0.46)	35.97 (±1.06)	36.44 (±0.80)	37.01 (±1.10)
MnO	0.0	0.0	0.0	0.0
MgO	2.11 (±0.61)	2.37 (±0.22)	4.35 (±0.77)	2.60 (±0.65)
CaO	0.22 (±0.03)	0.20 (±0.04)	0.06 (±0.02)	0.15 (±0.01)
Na_2O	0.24 (±0.08)	0.21 (±0.03)	0.06 (±0.04)	0.23 (±0.04)
K_2O	0.18 (±0.03)	0.14 (±0.02)	0.13 (±0.15)	0.12 (±0.03)
Total	84.29	86.92	88.95	89.22
Pseudo-1	$(Fe_{3.53}Mg_{0.37}Al_{1.74}Ca_{0.03}Na_{0.05}K_{0.03})(Si_{2.87}Al_{1.13})O_{10}(OH)_8$			
Cement-1	$(Fe_{3.37}Mg_{0.40}Al_{1.82}Ca_{0.02}Na_{0.05}K_{0.02})(Si_{2.91}Al_{1.09})O_{10}(OH)_8$			
Pseudo-2	$(Fe_{3.34}Mg_{0.71}Al_{1.71}Ca_{0.01}Na_{0.01}K_{0.02})(Si_{2.72}Al_{1.28})O_{10}(OH)_8$			
Cement-2	$(Fe_{3.38}Mg_{0.42}Al_{1.80}Ca_{0.02}Na_{0.05}K_{0.02})(Si_{2.89}Al_{1.11})O_{10}(OH)_8$			

Pseudo-1 and Cement-1 are the pseudomorphic aggregates and berthierine cement respectively in Area 1 (see *Pseudo (Be)* and *Be cmt* in Fig. 6d). Pseudo-2 and Cement-2 are the pseudomorphic aggregates and berthierine cement respectively in Area 2 (see *Pseudo (Be)* and *Be cmt* in Fig. 6c). The values in the table are the calculated weight percent of oxides based on microprobe analyses with the standard deviation (2σ) in parentheses. n = number of spots analyzed.

5c) surficially precipitated on (rather than within) the boundaries. This observation supports the notion that hematite postdates pressure solution.

Oolitic grains (Fig. 6a) are rare in the samples (< 0.2% of all grains, based on point counting). There is no evidence for extensive alteration of oolitic grains to hematite. Tourmaline grains have been previously characterized by microprobe analysis [35]. Tourmaline grains are present in different forms including rounded detrital grains, inclusions in detrital quartz grains, partially dissolved subangular grains and authigenic overgrowths (Fig. 6b). Most detrital tourmaline grains are dark brown and strongly pleochroic, reflecting a high Fe and Ti content [35]. Some detrital tourmaline grains have acicular light blue authigenic overgrowths. These authigenic tourmaline overgrowths (5–15 μm) hemimorphically nucleate on detrital tourmaline at the positive end of the *c*-crystallographic axis [35] and interfinger into the concentrically oriented laths of berthierine and illite (Fig. 6b).

We have also observed that some berthierine aggregates differ from pore-filling berthierine cements. The aggregates (Figs. 6c and d) have the following characteristics: (1) They have irregular outlines, (2) they are similar in size to adjacent quartz grains, and neatly fit into the framework of the quartz grains, (3) euhedral hematite flakes are distributed around these aggregates and are oriented perpendicular to aggregate boundaries, (4) contacts between quartz grains and these aggregates are sometimes sutured, and (5) quartz cement is present in the areas of berthierine cement, but absent in these berthierine aggregates. The above characteristics suggest that these berthierine aggregates are mineral pseudomorphs.

Chemical compositions and structural formulae of both pseudomorphic berthierine aggregates (Pseudo-1 and Pseudo-2) and pore-filling berthierine cement (Cement-1 and Cement-2) were calculated from the results of microprobe analyses (Table 2). The deficiency of the octahedral site probably reflects vacancies due to an excess of octahedral trivalent cations, especially Al. Electrical neutrality of the layers is achieved by vacant octahedral sites [36]. The presence of minor amounts of Ca, Na and K probably reflects contamination by other mineral phases. Although these elements have been included in the normalized formulae, their presence in the analyses has little effect on the structural formulae and the compositional arguments. Statistical tests at the 95% confidence level on the calculated weight percent of oxides reveal the differences in berthierine composition. Pseudo-1 is poorer in Al and Mg than Pseudo-2. Pseudo-2 is significantly Mg richer and Si poorer than Cement-2 whereas Pseudo-1 and Cement-1 are statistically the same. In addition, there is no significant difference between Cement-1 and Cement-2, especially for Fe and Mg contents.

4. Interpretation and discussion

Petrographic observations (optical, SEM and XRD) in conjunction with quantitative electron microprobe analyses show the textural and compositional relationships between the different minerals associated with hematite authigenesis. Deformation features are concentrated in the detrital quartz grains (Fig. 4a) whereas hematite–berthierine cements and pseudomorphed detrital grains are undeformed (Figs. 4b, 6c and 6d). Pressure solution results in pore size reduction and an increase in the contact area between detrital grains. If the rock was cemented prior to pressure solution, there must be some accommodation in the cements for volume reduction and removal of cements. Therefore, the absence of the pressure solution seams (Fig. 5a) and absence of deformation (Fig. 5b) in the hematite–berthierine–quartz cements indicate that hematite and berthierine precipitated after pressure solution due to deep-burial compaction and/or layer-parallel shortening.

The euhedral habits of hematite and berthierine cements in the pore spaces indicate a diagenetic origin [23]. The formation and survival of the pore-lining hematite cement is an indication of oxidizing conditions during cementation. Euhedral hematite flakes perpendicular to detrital grain boundaries (Fig. 4d) reflect growth of hematite into a fluid-filled open void. The ran-

domly oriented berthierine crystals in the interior of the pore space (Fig. 4b) show that berthierine formed no earlier than hematite. Paleomagnetic studies show that the carrier of remagnetization is hematite and the paleopole is Late Paleozoic in age and probably synfolding. Peak temperatures in these rocks were $< 250°C$ [35], and therefore the remagnetization is probably a CRM acquired at the time when pore-lining hematite formed. In addition, multiple intergrowths of berthierine and hematite (Fig. 4d) suggest that the formation of the two minerals is broadly coeval. Although it has been reported that berthierine and chamosite in the Clinton oolitic ironstones is depositional [e.g., 28,29,37,38], the intergrowth textures (Figs. 4d and 6c) and timing of remagnetization residing in hematite indicate that the cements of hematite, berthierine and quartz are syn-Alleghanian diagenetic precipitates.

The berthierine aggregates (Figs. 6c and d) are interpreted as mineral pseudomorphs. Processes of pseudomorphism could include (1) replacement of precursor mineral grains and or (2) dissolution of precursor grains and later precipitation of other mineral species. The occurrence of sand-size aggregates of diagenetic layer silicates fitting tightly into a stable framework of detrital grains suggests pseudomorphic replacement after unstable silicate grains [23]. The same berthierine texture (randomly oriented laths) in both pore-filling cement and berthierine aggregates indicates that these berthierines are possibly products of the same diagenetic event. Differing berthierine compositions of some pseudomorphs (e.g., Pseudo-2) with respect to others (e.g., Pseudo-1) and also with respect to cements (e.g., Cement-2) suggests local control by the composition of the precursor detrital grains. The composition of Pseudo-2 differing from Pseudo-1 and Cement-2 may reflect the original composition of the detrital Fe-bearing silicates. In contrast, the overall uniform composition in the berthierine cement indicates that the cementation was controlled by externally derived fluids.

The Fe-rich composition of the pseudomorphs supports our interpretation that the precursor minerals were detrital Fe-Mg silicates as suggested previously by Folk [30]. Fe-bearing silicates were generally unstable in an oxidizing post-depositional environment and continued to disintegrate or dissolve during intrastratal alteration [39,40]. These detrital Fe-bearing silicates could have provided a significant proportion of Fe for the later hematite and berthierine cements [30]. One possible explanation for the present scarcity of Fe-bearing silicates is that they have been dissolved or converted to hematite and berthierine during diagenesis. Some detrital tourmaline grains are still present because they are much more stable than other detrital Fe-Mg silicates (e.g., amphiboles, pyroxenes). The greater stability of tourmaline during diagenesis is also indicated by overgrowths of tourmaline fibers (Fig. 6b). Hence the detrital tourmaline grains may represent the diagenetic survivors of the original population of ferromagnesian grains.

Alternatively, hematite authigenesis could be related to late diagenetic alteration of initial chamositic clay minerals (or berthierine) in oolitic grains [28,29]. This mechanism is not applicable to our samples because the oolitic grains are rare ($< 0.2\%$). It has also been proposed that Late Paleozoic authigenic hematite was derived from berthierine or Fe-chlorite (chamosite) cement, which formed during early diagenetic ferruginization [28]. Although this is a possible origin of hematite authigenesis, the pore-lining nature of Alleghanian hematite cementation suggests that early Fe-bearing cements must have undergone complete dissolution before later authigenic hematite and berthierine precipitated from solution. Furthermore, the possibility that the precursor of hematite was an amorphous Fe-hydroxide cannot be completely excluded. Early goethite may have been converted to hematite during subsequent deep burial diagenesis [41,42].

Based on the Fe content in the berthierine from microprobe analyses and estimated hematite (10%) and berthierine (11%) contents from the image analyses, we performed a source–sink mass balance calculation assuming constant volume. Considering the extreme case in which all Fe is from detrital Ca-Fe amphibole (e.g., ferroactinolite) and Ca-Fe pyroxene (hedenbergite), this would require that more than 40% of the original grains be detrital Fe-Mg silicates. However, the

present content of detrital quartz grains (69%) determined from image analysis implies that Fe derived from local detrital Fe-bearing silicate grains alone is not sufficient to account for all of the authigenic hematite and berthierine observed in the samples. An external Fe source is needed to produce the observed amount of hematite. Two end-member scenarios for the introduction of Fe are diffusion in a static fluid medium and transport in a moving fluid. The high porosities of these rocks (> 20%) prior to hematite–berthierine cementation could have permitted diffusion of Fe over distances of > 10 m in 10,000 yr [43]. Additional Fe may have been released from the stratigraphically adjacent shales during smectite–illite transformation as a consequence of burial. Alternatively, high porosities also suggest high permeability, and therefore Fe could have been introduced by fluid flow. Further work over a larger area is needed to evaluate the relative influence of these possibilities.

The origin of hematite in red beds varies with depositional environments and diagenetic conditions. The characteristics of hematite authigenesis in red beds depend on a variety of factors, such as lithology, geochemical conditions, porosity, permeability and interaction between fluids and the rocks. Different members of the same formation could have different diagenetic histories [30]. Our interpretations are based solely on the observations of a hematite–berthierine-cemented quartz sandstone member of the Rose Hill Formation. The mechanism proposed in this paper does not negate observations from other lithologies in the Rose Hill Formation.

5. Conclusions

A Late Paleozoic chemical remagnetization residing in hematite is observed in a quartz sandstone bed in the Silurian Rose Hill Formation of the Central Appalachians. Comparison of paleopole positions of the Rose Hill Formation with Late Paleozoic poles of North America suggests synfolding remagnetization acquired during the Alleghanian. Pore-lining hematite, pore-filling berthierine and quartz cements postdate pressure solution of the detrital quartz grains. Authigenic hematite and berthierine are interpreted to be syn-Alleghanian precipitates. Pseudomorphic replacement of berthierine aggregates after detrital Fe-Mg silicate grains indicates an important Fe source for hematite and berthierine cements, although an external Fe source is also required.

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